

A Synopsis

On

BuildMyWebsiteAI

*An Autonomous AI-Powered Multi-Page Web Application Generator Platform with Dynamic Theme Engine
& Real-Time Gemini AI Chatbot Assistant*

of

Mini PROJECT (CA-353)

MCA-II Year / III Semester

Submitted to the Department of Computer Application by:

1. Ayush Chaubey (Team Leader)	Roll No: 2501640140028
2. Dev Tripathi (Team Member)	Roll No: 2501640140038
3. Ritesh Mishra (Team Member)	Roll No: 2501640140083
4. Ayush Tiwari (Team Member)	Roll No: 2501640140031
5. Yashraj Dixit (Team Member)	Roll No: 2501640140113

Under the Supervision of:

Dr. Narendra Kumar Sharma

(Designation, Department of Computer Application)

PRANVEER SINGH INSTITUTE OF TECHNOLOGY, KANPUR

Approved by AICTE and Affiliated to

Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW

Academic Session 2026–2027

2. TABLE OF CONTENT

S.No.	Topic / Section Title	Page No.
1.	Front Page & Team Metadata	1
2.	Table of Content	2
3.	Certificate & Declaration	3
4.	INTRODUCTION	4
5.	Gap in Study	5
6.	Proposed System Architecture	6
7.	MINI PROJECT SCOPE	7
8.	AIMS AND OBJECTIVE	8
9.	Brief Modules Description	9
10.	Tools and Technology Used	10
11.	System Requirement (Hardware & Software)	11
12.	METHODOLOGY (Frontend, Backend, DFD & ERD)	12
13.	EXPECTED TIME SCHEDULE (Gantt Chart)	13
14.	IMPACT OF PROPOSED SYSTEM IN ACADEMICS & INDUSTRY	13
15.	ROLES AND RESPONSIBILITY	14
16.	PROS AND CONS	14
17.	CONCLUSION AND FUTURE SCOPE	15
18.	REFERENCES	15

3. DECLARATION & CERTIFICATE

DECLARATION

We hereby declare that the Mini Project Synopsis entitled "**BuildMyWebsiteAI: An Autonomous AI-Powered Multi-Page Web Application Generator Platform**" submitted in partial fulfillment of the requirements for the award of the degree of **Master of Computer Application (MCA)** to **Pranveer Singh Institute of Technology, Kanpur** affiliated to **Dr. A.P.J. Abdul Kalam Technical University, Lucknow** comprises authentic research work carried out by us under guidance. Due acknowledgement has been made in the text to all other materials used.

CERTIFICATE

This is to certify that Report entitled "**BuildMyWebsiteAI**" which is submitted by **Ayush Chaubey (Roll No: 2501640140028)**, **Dev Tripathi (Roll No: 2501640140038)**, **Ritesh Mishra (Roll No: 2501640140083)**, **Ayush Tiwari (Roll No: 2501640140031)**, and **Yashraj Dixit (Roll No: 2501640140113)** in partial fulfillment of the requirement for the award of degree Master of Computer Application to **Pranveer Singh Institute of Technology, Kanpur** Dr. A P J A K Technical University, Lucknow is a record of the candidates own work carried out by them under my supervision. The matter embodied in this thesis is original and has not been submitted for the award of any other degree.

Date: _____

Signature: _____

Dr. Narendra Kumar Sharma

(Supervisor, Department of MCA)

Approved By: _____

Head of Department (HOD)

Official Stamp / Seal:

PSIT Kanpur

4. INTRODUCTION

In the contemporary digital era, establishing a high-performance web presence is vital for businesses, startups, educational institutions, and professionals. However, traditional website development involves lengthy software development life cycles (SDLC), requiring hand-coded HTML/CSS/JS frameworks, responsive breakpoints, server configurations, and database integrations. While low-code/no-code platforms like Wix, Squarespace, and WordPress simplified website building, they suffer from steep learning curves, rigid design templates, heavy code bloat, and subscription lock-in.

BuildMyWebsiteAI bridges this gap by pioneering an autonomous artificial intelligence platform that generates production-grade, fully responsive multi-page web applications directly from natural language prompts. Operating on a modern decoupled architecture powered by **React 18, Vite, TailwindCSS, FastAPI, SQLAlchemy, MySQL, and Google Gemini 3.5 AI**, BuildMyWebsiteAI constructs multi-page page tree structures, curates semantic typography and palette tokens, renders interactive section layouts with AOS scroll animations, and provides 1-click HTML/CSS ZIP bundle exports.

Furthermore, the platform incorporates a **Global Dark ■ / Light ■■ Theme Engine**, seamless user authentication with JWT security, an interactive **Super Admin Control Portal**, and a dedicated **Floating AI Chatbot Assistant** connected live to the Google Gemini 3.5 Flash engine capable of dynamically answering any technical or design query in real time.

5. GAP IN STUDY

- **Single-Page Limitations vs Multi-Page Demands:** Most existing AI site generators create only a single landing page, failing to structure authentic multi-page websites (`Home`, `About Us`, `Services`, `Showcase`, `Pricing`, `Contact Us`).
- **Code Bloat & Proprietary Lock-In:** Traditional drag-and-drop website builders produce convoluted, unoptimized code output that cannot be easily exported or self-hosted independently.
- **Static Theme Constraints:** Existing AI tools lack unified, global surface-color synchronization when toggling between Dark and Light mode across all dashboard cards, user tables, and canvas views.
- **Absence of Embedded AI Mentorship:** Web builders lack an integrated, real-time AI Assistant capable of providing prompt engineering advice, category guidance, and platform answers directly within the workspace.
- **Deployment & Export Barriers:** Platforms obscure generated code files behind paywalls instead of providing 1-click clean HTML/CSS/JS ZIP archive downloads and co-domain

network publishing.

6. PROPOSED SYSTEM

- **Autonomous Page Tree Generator:** Converts natural language user prompts into detailed JSON Page Tree schemas containing structured section properties for 18 industry verticals (AI Startup, SaaS, E-Commerce, Healthcare, Real Estate, Restaurant, etc.).
- **Dedicated Multi-Page Router:** Implements full client-side SPA routing (`/`, `/features`, `/showcase`, `/how-it-works`, `/pricing`, `/faq`, `/dashboard`, `/admin`) backed by static Render fallback rewrite rules (`/\* /index.html 200`).
- **Universal Dark ■ / Light ■■ Theme Engine:** Features a single circular theme toggle in the top navbar that dynamically updates CSS variable tokens, surface cards, metrics panels, and HTML5 Canvas particle background shaders globally.
- **Interactive Mouse Laser Particle Background:** An HTML5 Canvas rendering engine featuring particle magnetic attraction physics, energy laser connection beams, and cursor trail motion ripples across 100% of pages.
- **Real-Time Gemini 3.5 AI Chatbot Assistant:** A floating global AI assistant widget powered by Google Gemini 3.5 REST API capable of answering any technical, design, or prompting query live.
- **Robust Export & Deployment Engine:** Generates standard W3C-compliant standalone HTML5/CSS3 ZIP archives equipped with AOS scroll animation CDN scripts and instant 1-click co-domain network publishing.

7. MINI PROJECT SCOPE

- **Functional Scope:** Covers user registration, OTP email verification, JWT authentication, 18 category website generation, multi-page routing, live desktop/tablet/mobile viewport canvas preview, 1-click ZIP export, co-domain deployment, and super admin management.
- **Non-Functional Scope:** Guarantees sub-second API response times, 100% viewport responsiveness across all device form factors, HTTPS SSL data encryption, and resilient JSON deserialization (`safe\_dict`).
- **Target Audience:** Small business owners, tech startups, freelancers, digital agencies, students, and non-technical entrepreneurs seeking instant professional websites.

8. AIMS AND OBJECTIVE

- **Primary Aim:** To engineer an autonomous AI-driven platform that reduces web development time from days to under 10 seconds while maintaining professional aesthetic quality.

- **Objective 1:** Implement Google Gemini 3.5 AI REST API integration for dynamic text content generation and conversational chatbot mentorship.
- **Objective 2:** Build a responsive client-side SPA architecture using React 18, Framer Motion, and TailwindCSS with zero layout distortion.
- **Objective 3:** Construct a secure RESTful API backend using FastAPI, Pydantic, SQLAlchemy, and MySQL hosted on Aiven Cloud.
- **Objective 4:** Ensure 100% self-contained code output through downloadable ZIP archives containing clean HTML, CSS, and JS files.

9. BRIEF MODULES DESCRIPTION

• **Authentication & Access Control Module:** Manages user registration, bcrypt password hashing, OTP verification via SMTP Gmail, JWT bearer token generation, and role-based access control (User vs Super Admin).

• **AI Prompt & Page Tree Generation Module:** Interprets natural language prompts, selects optimal category design tokens (colors, typography, hero media), and constructs structured multi-page JSON page tree trees.

• **Canvas Preview & Multi-Page Router Module:** Renders live web layouts inside desktop, tablet (768px), and mobile (375px) viewports with real-time sub-page navigation (`Home`, `About Us`, `Services`, `Contact Us`).

• **Export & Deployment Engine Module:** Converts database page tree schemas into W3C-compliant standalone HTML5 files and CSS stylesheet bundles, packaging them into downloadable ZIP archives.

• **Super Admin Portal Module:** Provides platform-wide metrics overview (total users, active projects, live co-domains), user management table, password reset capabilities, and 1-click user session impersonation.

• **Google Gemini 3.5 AI Chatbot Module:** A global floating widget connected directly to Google Gemini 3.5 REST API models (`gemini-3.5-flash`), featuring conversation history memory and custom API key override settings.

10. TOOLS AND TECHNOLOGY USED

Layer / Category	Technology / Framework	Purpose & Role
Frontend Core	React 18, Vite, JavaScript (ES6+)	High-performance SPA rendering & fast build pipeline
Styling & UI	TailwindCSS, Vanilla CSS, Lucide Icons	Utility-first styling, glassmorphism, responsive layout
Animations	Framer Motion, HTML5 Canvas API	Interactive particle laser physics & dynamic transitions
Backend Core	Python 3.13, FastAPI, Uvicorn	Asynchronous RESTful API server & request routing
Database & ORM	MySQL (Aiven Cloud), SQLAlchemy, PyMySQL	Relational data storage, ORM models, connection pooling

AI Intelligence	Google Gemini 3.5 Flash REST API	Natural language page tree generation & live AI Chatbot
Security & Mail	PyJWT, Passlib (Bcrypt), SMTPLib	JWT token auth, password hashing, OTP verification
Deployment	Render Cloud, GitHub, Git	Automated CI/CD production hosting & version control

11. SYSTEM REQUIREMENT

11.1 Hardware Requirements:

- **Processor:** Intel Core i3 / AMD Ryzen 3 or higher (Quad-core recommended).
- **System Memory (RAM):** Minimum 4 GB (8 GB recommended for concurrent dev servers).
- **Storage Space:** Minimum 2 GB free disk space (SSD recommended).
- **Network:** Broadband Internet Connection (Minimum 5 Mbps for AI API calls & CDN scripts).

11.2 Software Requirements:

- **Operating System:** Windows 10/11, macOS 12+, or Ubuntu Linux 20.04+.
- **Runtime Environment:** Node.js v18.0+ & Python 3.10+ (Python 3.13 tested).
- **Database Server:** MySQL 8.0+ (Localhost or Aiven Cloud Database instance).
- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge, or Apple Safari.

12. METHODOLOGY

12.1 Frontend Methodology to Implement Mini Project:

The frontend methodology relies on a component-driven Single Page Application (SPA) architecture using React 18 and Vite. All visual components consume global theme tokens from a central ``ThemeContext.jsx``. Page routing is managed by ``react-router-dom`` with ``_redirects`` rewrite rules (``/* /index.html 200``) ensuring seamless client-side page refreshes on static hosts. An HTML5 Canvas element (``AnimatedBackground.jsx``) tracks cursor coordinates ``(mouseX, mouseY)`` to calculate particle vector forces, render energy laser connection beams, and draw glowing mouse trail ripples dynamically.

12.2 Backend Methodology to Implement Mini Project:

The backend methodology adopts a layered microservices pattern built on FastAPI and SQLAlchemy. Request workflows follow: Router -> Dependency Interceptor (JWT Verification) -> Controller -> Service Layer -> Database ORM. To prevent runtime crashes during MySQL data retrieval, the backend implements a ``safe_dict`` utility that automatically parses JSON strings and safely handles ``NoneType`` sections. The export pipeline utilizes Python's ``zipfile`` and ``io.BytesIO`` modules to construct W3C-compliant standalone HTML5 files and CSS stylesheets in-memory before streaming binary blobs to the client.

12.3 System Design (Data Flow Diagram & ERD):

Data Flow Diagram (DFD Level 0): User sends prompt -> Frontend API client transmits JSON payload -> FastAPI Router validates JWT -> Gemini 3.5 AI Engine generates Page Tree -> SQLAlchemy persists Website entity in MySQL -> Frontend renders live Canvas Preview.

Entity-Relationship Diagram (ERD):

- **User Entity:** ``id`` (PK), ``full_name``, ``email`` (Unique), ``hashed_password``, ``is_verified``, ``is_admin``, ``created_at``.
- **Website Entity:** ``id`` (PK), ``user_id`` (FK -> User.id), ``title``, ``category``, ``subdomain``, ``theme``, ``page_tree`` (JSON Text), ``is_published``, ``created_at``.
- **Relationship:** One-to-Many (1 User owns N Websites).

13. EXPECTED TIME SCHEDULE (Gantt Chart)

Phase / Project Activity	Duration	Status
Phase 1: Requirement Analysis & System Architecture Design	Weeks 1 - 2	Completed

Phase 2: Database Schema (MySQL/Aiven) & Auth API (JWT/OTP)	Weeks 3 - 4	Completed
Phase 3: AI Prompt Generator Engine & Gemini 3.5 API Integration	Weeks 5 - 7	Completed
Phase 4: Frontend Multi-Page SPA, Theme Engine & Canvas Preview	Weeks 8 - 10	Completed
Phase 5: ZIP Export Engine, Co-Domain Deploy & Gemini AI Chatbot	Weeks 11 - 13	Completed
Phase 6: Super Admin Control Panel, System Testing & Render Deployment	Weeks 14 - 15	Completed

14. IMPACT OF PROPOSED SYSTEM IN ACADEMICS AND INDUSTRY

Academic Impact: Serves as an exemplary benchmark project illustrating modern decoupled full-stack architecture, LLM API orchestration, real-time HTML5 Canvas animation physics, and database resilience pattern (`safe\_dict`). It provides a foundation for future research in automated code synthesis and AI-assisted UX engineering.

Industry Impact: Democratizes web development for small enterprises, startups, non-profits, and educational bodies by reducing website creation costs to zero while delivering production-ready, exportable HTML/CSS code without vendor lock-in.

15. ROLES AND RESPONSIBILITY

Team Member / Role	Assigned Responsibilities
Ayush Chaubey (Team Leader / Full-Stack & AI Lead)	Full-stack SPA design (React 18/Vite), FastAPI backend server, Gemini 3.5 AI Integration, Super Admin Portal, Deployment.
Dev Tripathi (Team Member / Backend & DB)	MySQL Database schemas (Aiven Cloud), SQLAlchemy ORM models, JWT auth security & bcrypt hashing.
Ritesh Mishra (Team Member / Frontend & UI)	TailwindCSS design tokens, glassmorphism theme components, responsive viewport preview handlers.
Ayush Tiwari (Team Member / Export & Animations)	HTML5 Canvas magnetic particle laser background shader, W3C standalone ZIP code export engine.
Yashraj Dixit (Team Member / QA & Testing)	System route verification, API endpoint validation, CORS header security, manual viewport QA testing.

16. PROS AND CONS

16.1 System Advantages (Pros):

- **Instant Multi-Page Generation:** Produces structured 4-page websites in under 10 seconds from a single prompt.

- **Zero Code Lock-In:** 1-click ZIP downloads containing clean HTML5, CSS3, and JS files ready for self-hosting.
- **Dynamic Gemini 3.5 AI Assistant:** Global chatbot mentorship available on every screen for real-time prompt guidance.
- **Universal Dark/Light Mode:** Synchronized card and background palette toggling across all dashboard views.

16.2 System Constraints (Cons):

- **Internet Dependency:** Requires active network connectivity to reach Google Gemini REST APIs and Aiven MySQL.
- **API Rate Limits:** Dependent on Gemini API quota limits for high-volume concurrent prompt generations.

17. CONCLUSION AND FUTURE SCOPE

Conclusion: BuildMyWebsiteAI successfully demonstrates the power of autonomous AI in web software engineering. By integrating React 18, FastAPI, MySQL, and Google Gemini 3.5 AI, the platform delivers a production-grade multi-page web builder equipped with responsive viewports, ZIP code exports, dynamic theme conversion, and real-time AI assistance.

Future Enhancements:

- Integration with custom domain DNS mapping and automated SSL certificate provisioning.
- Drag-and-drop visual component editing directly within the live preview canvas.
- Multi-language AI translation engine supporting 20+ global languages.

18. REFERENCES

[1] FastAPI Documentation, *FastAPI Modern Python Web Framework*, <https://fastapi.tiangolo.com/>

[2] React Documentation, *React 18 Component Architecture*, <https://react.dev/>

[3] Google Gemini API Documentation, *Gemini 3.5 Flash REST API Reference*, <https://ai.google.dev/docs>

[4] SQLAlchemy Documentation, *SQLAlchemy ORM Database Toolkit for Python*, <https://www.sqlalchemy.org/>

[5] TailwindCSS Documentation, *Utility-First CSS Framework*, <https://tailwindcss.com/>

[6] W3C Web Content Accessibility Guidelines (WCAG 2.1), *World Wide Web Consortium*,
<https://www.w3.org/TR/WCAG21/>